

Software Engineering Task 2 Folio

William Lam

Requirements Definition

Functional

- Remote control with two buttons that open and close the garage door
- Use at least 3 sensors and actuators
- Lights automatically turn on when a car enters the garage
- Has a manual mode to manually open the garage door
- Door automatically closes after the car enters the garage
- Ultrasonic sensor that makes the microbit beep if a car is too close
- LED display to show ENTER and LEAVE on the microbit

Nonfunctional

- Reliability: Must be able to consistently perform its functions
- Safety: Must be electrically safe from surges and shocks
- Power Efficiency: Must be able to run off two battery packs maximum
- Precision: All data collected should be accurate. Additionally all actuators should work without fault
- Durability: Must survive being moved around
- Usability: Must be intuitive as well as easy to reprogram

Determining Specifications

Component List

- 2 x Microbits
- Ultrasonic sensor
- Bitmaker Expansion Board
- Servo
- Rotary Sensor
- Neopixels
- 2 x Battery Packs

Dimensions

24.5cm (Length) x 16cm (Width) x 10.5cm (Height)

List of sensors and Actuators

Sensors:

- Ultrasonic
- Buttons A and B
- Rotary
- Touch
- Radio transmitter

Actuators:

- 5x5 LED screen
- Speaker
- Radio transmitter
- Servo
- Neopixels

Power Supplies

The first microbit (remote control) will be powered by a battery pack of 2 AAA batteries.

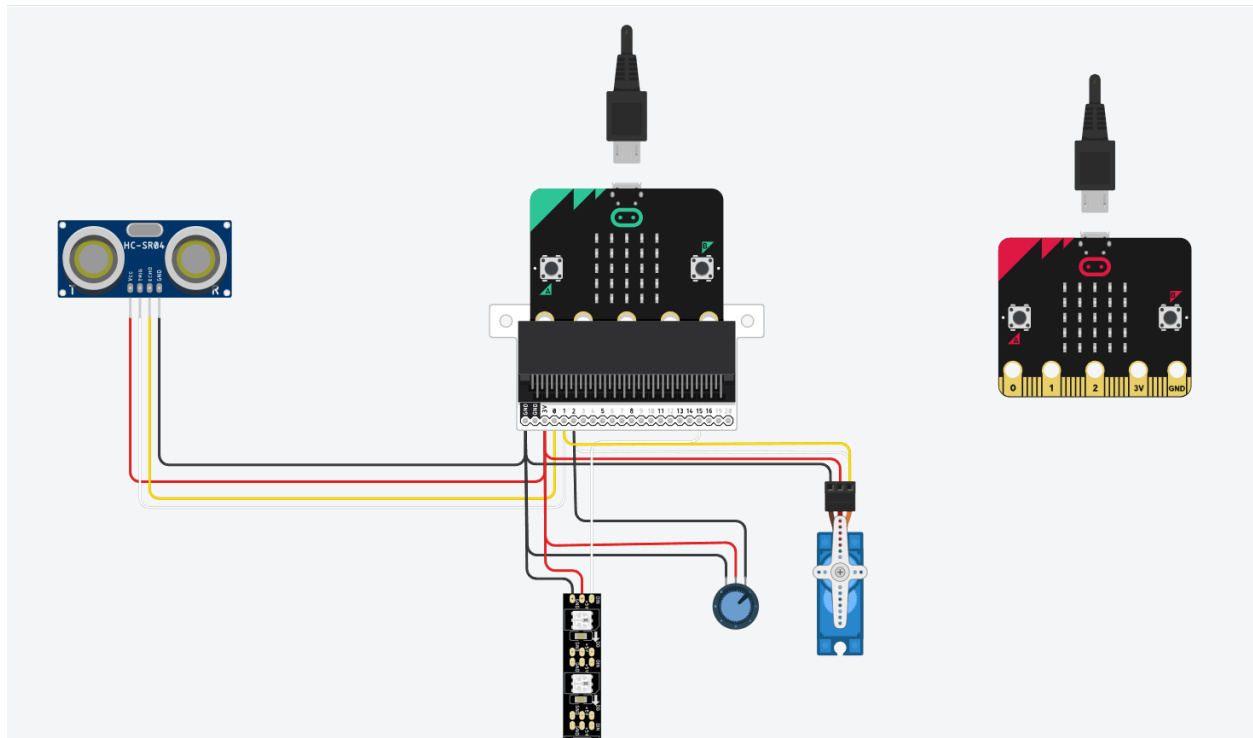
The second microbit will be powered by a bitmaker expansion board. Its Bitmaker Expansion Board will be powered by a 2000mah power bank.

Design

Component List

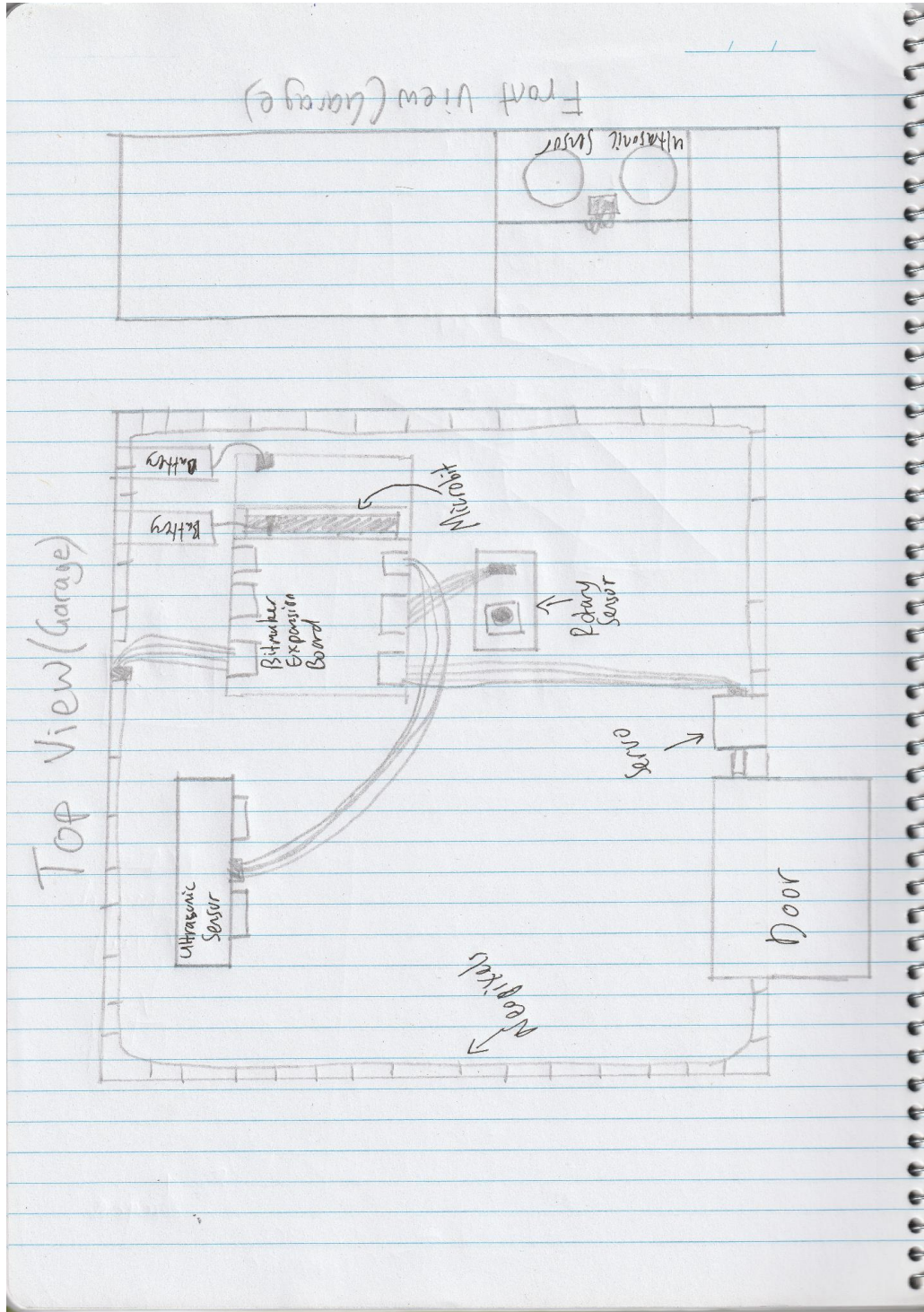
- 2 x Microbits
- Ultrasonic sensor
- Bitmaker Expansion Board
- Servo
- Rotary Sensor
- Neopixels
- 2 x Battery Packs

Wiring Circuit



Development

Sketch of Prototype



Journal

Date	Objective	Description	For Next Time
22/06/2025	Begin planning	Folio was created and notification was viewed to create a list of what to do	Begin preliminary research
28/06/2025	Begin preliminary research	- Basic notes were taken - Requirements definition and determining specifications completed - Wiring Diagram was created - Design was started	Finish Design and begin Development
29/06/2025	Finish Design and begin development	- Design was continued, was not finished	Finish Design and begin development
30/06/2025	Finish Design and begin development	Requirements definition, determining specifications and design were restarted and completed to better suit the marking criteria. Code was started and finished Modules of the code: Set_servo_angle: calculates the duty cycle for the servo Get_distance: code for the ultrasonic sensor to calculate distance of an object Manual_mode: allows the user to manually open the door Enter_garage: what is run when a car enters the garage (door opens, lights turn on, car enters, door closes) Close_garage: what is run when a car leaves the garage (door opens, car leaves, door closes, lights off)	Do sketches and begin building a prototype

		Main: main program which receives radio messages and enters manual mode	
01/07/2025	Do sketches and begin building a prototype	Paper sketch of the prototype was completed A prototype was built and code was debugged accordingly. Added two more functions: Lights() Doors() All main recording was completed and video editing began	Finish video, go over folio and submit
02/07/2025	Finish video, go over folio and submit	Some extra recording needed to be completed and was done swiftly. Video was being edited and finalised. Evaluation was completed Folio submitted	NIL

Algorithms and Working Code

Pseudocode

Remote Control:

```

FROM microbit IMPORT all

IMPORT radio

CONFIGURE radio with group 2
TURN ON radio

FUNCTION button_A_pressed():
    SEND radio 'ENTER'
    SCROLL 'ENTER'
    CLEAR Display
    RETURN
ENDFUNCTION

FUNCTION button_B_pressed():

```

```

        SEND radio 'LEAVE'
        SCROLL 'LEAVE'
        CLEAR Display
        RETURN
ENDFUNCTION

FUNCTION main()
    IF button A pressed THEN
        CALL button_A_pressed()
    ELIF button B pressed THEN
        CALL button_B_pressed()
    ENDIF
ENDFUNCTION

WHILE True
    Call main()
ENDWHILE
while True:
    main()

```

Garage:

```

FROM microbit IMPORT all

IMPORT radio
IMPORT neopixel
IMPORT time
IMPORT music

INITIALIZE neopixels on pin15 with 30 LEDs
CLEAR all neopixels
SET lights_on TO 0
SET door_open TO 0

CONFIGURE radio with group 2
TURN ON radio

CONFIGURE servo on pin2 with 20ms analog period
SET open_angle TO 100
SET close_angle TO 35

SET too_close TO 3
SET too_far TO 15

FUNCTION SetServoAngle(angle)
    IF angle >= 10 AND angle <= 170 THEN
        SET duty TO 25.6 + (angle × 0.57)
        WRITE duty TO pin2 (servo)
        RETURN duty
    ENDIF
END FUNCTION

FUNCTION GetDistance()
    SET _TIMEOUT1 TO 1000
    SET _TIMEOUT2 TO 10000

    SEND 10µs pulse on pin0 to trigger ultrasonic sensor

    SET count TO 0
    WHILE count < _TIMEOUT1 AND pin0 is LOW
        INCREMENT count
    ENDWHILE
    SET t1 TO current time in microseconds

    SET count TO 0
    WHILE count <= _TIMEOUT2 AND pin0 is HIGH
        INCREMENT count
    ENDWHILE
    SET t2 TO current time in microseconds

    SET duration TO t2 - t1
    CALCULATE distance = (duration × 0.0343) ÷ 2
    RETURN INTEGER value of distance
END FUNCTION

FUNCTION lights():
    GLOBAL lights_on
    PRINT str(lights_on)
    IF lights_on == 1 THEN

```

```

        CLEAR neopixels
        SET lights_on TO 0
    ELSE
        SET neopixels to blue
        SHOW neopixels
        STOP 200
        SET lights_on TO 1
    ENDIF
    RETURN
ENDFUNCTION

FUNCTION door():
    GLOBAL door_open
    PRINT str(door_open)
    IF door_open == 0 THEN
        CALL SetServoAngle(open_angle)
        SET door_open TO 1
    ELSE
        CALL SetServoAngle(close_angle)
        SET door_open TO 0
    ENDIF
    RETURN
ENDFUNCTION

FUNCTION ManualMode()
    WHILE logo pin is NOT touched
        IF logo pin is touched THEN
            PLAY tone at 600Hz
            WAIT 200ms
            STOP tone
            WAIT 3 seconds
            RETURN
        ENDIF

        READ analog value from rotary dial on pin1 INTO turn
        MAP turn to angle between close_angle and open_angle
        CALL SetServoAngle(turn)

        WAIT 2 seconds

        PRINT turn and pin2 analog value

        IF logo pin is touched THEN
            PLAY tone at 600Hz
            WAIT 200ms
            STOP tone
            WAIT 3 seconds
            RETURN
        ENDIF
    ENDWHILE
END FUNCTION

FUNCTION EnterGarage()
    GLOBAL lights_on
    GLOBAL door_open

    CALL SetServoAngle(open_angle)
    SET neopixels to blue
    SHOW neopixels
    SET lights_on TO 1

    SET car_distance TO GetDistance()
    WHILE car_distance > too_close
        SET car_distance TO GetDistance()
    ENDWHILE

    IF car_distance <= too_close THEN
        PLAY warning tone at 440 Hz
        WAIT 200ms
        STOP tone
    ENDIF

    WRITE LOW to pin0
    WAIT 1ms

    WAIT 2 seconds

```



```

        CALL SetServoAngle(close_angle)
        SET door_open TO 0
        RETURN
    END FUNCTION

FUNCTION LeaveGarage()
    GLOBAL lights_on
    GLOBAL door_open

    CALL SetServoAngle(open_angle)
    SET door_open TO 1

    SET car_distance TO GetDistance()
    WHILE car_distance < too_far
        SET car_distance TO GetDistance()
    ENDWHILE

    IF car_distance >= too_far THEN
        PLAY tone at 440 Hz
        WAIT 200ms
        STOP tone
    ENDIF

    WRITE LOW to pin0
    WAIT 1ms

    WAIT 2 seconds

    CALL SetServoAngle(close_angle)
    SET door_open TO 0

    CLEAR neopixels
    SET lights_on TO 0
END FUNCTION

FUNCTION Main()
    GLOBAL lights_on
    IF logo pin is touched THEN
        DISPLAY image M
        PLAY tone at 600Hz
        WAIT 200ms
        STOP tone

        SET neopixels to yellow
        SHOW neopixels
        WAIT 2 seconds

        CALL ManualMode()

        CLEAR neopixels and display
        RESET micro:bit (stop ultrasonic noise)

    ENDIF

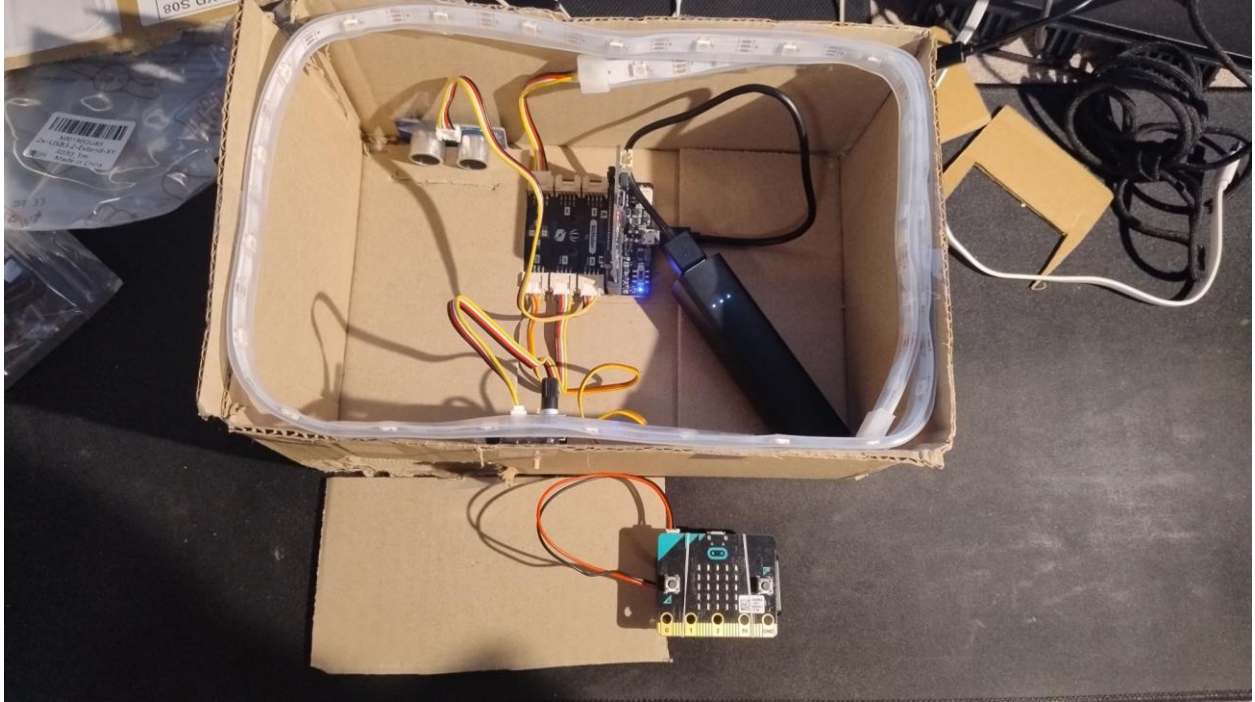
    SET message TO radio.receive()
    IF message = "CLOSE" THEN
        CALL LeaveGarage()
    ELSEIF message = "OPEN" THEN
        CALL EnterGarage()
    ENDIF
END FUNCTION

WHILE TRUE
    CALL Main()
ENDWHILE

```

Prototype

Photograph



How it Works

Two Main Components:

- Remote Control
- Garage

Remote Control

Button A pressed: Sends a radio message to the garage to begin the enter_garage function. It then displays 'ENTER' on the LED screen.

Button B pressed: Sends a radio message to the garage to begin the leave_garage function. It then displays 'LEAVE' on the LED screen.

Garage

When powered on, it constantly looks for radio messages. When it receives a message it goes into one of two functions:

- enter_garage
- leave_garage

Enter_garage

The door will open, lights will turn on to blue, and the ultrasonic sensor will begin calculating the distance of the car. As the car approaches the sensor, the distance closes. Once it reaches a certain distance, the microbit will beep, telling the car to stop. It will then automatically close the door.

Leave_garage

The door will open, and the ultrasonic sensor will calculate the distance of the car as it leaves the garage. Once the car has fully exited, the microbit will beep, and the door will close. The lights will then turn off.

When it is still searching, the buttons and the logo can be pressed.

Button A pressed: Acts as a toggle for the lights (neopixels).

Button B pressed: Acts as a toggle for the door (servo).

Logo pressed: Activates manual_mode.

Manual_mode

After the logo has been pressed, the lights will turn to green and the microbit will beep. An 'M' will be displayed on the LEDs to signify manual mode. The rotary sensor can now be used as a knob for opening and closing the door. The reading from the rotary sensor is converted into an integer angle to be passed onto the servo which then moves the door. The formula is:

Turning Angle = Rotary Sensor Value * (open_angle - close_angle) / 1021 + close_angle)

To leave manual mode, press the logo again. After a short delay it will beep, turn off the green lights, and clear the LED display, signalling it has left manual mode.

Evaluation

Criteria	Result
- Remote control with two buttons that open and close the garage door	Passed
- Use at least 3 sensors and actuators	Passed
- Lights automatically turn on when a car enters the garage	Passed
- Has a manual mode to manually open the garage door	Passed
- Door automatically closes after the car enters the garage	Passed
- Ultrasonic sensor that makes the microbit beep if a car is too close	Passed
- LED display to show ENTER and LEAVE on the microbit	Passed
- Reliability: Must be able to consistently perform its functions	Passed
- Safety: Must be electrically safe from surges and shocks	Passed
- Power Efficiency: Must be able to run off two battery packs maximum	Passed
- Precision: All data collected should be accurate. Additionally all actuators should work without fault	Passed
- Durability: Must survive being moved around	Passed
- Usability: Must be intuitive as well as easy to reprogram	Passed

Positive:

My system filled in all the specified criteria and worked completely fine consistently. It achieves the goal of automating otherwise time consuming tasks, allowing the user to quickly enter and exit the garage.

The system is easily assembled and disassembled, only requiring common household items such as cardboard and tape. It can easily be moved around due to its compact size and shape as a rectangle, meaning it is easy to grip and control.

There is a physical manual mode, which can prove useful in scenarios where the automated system fails (One such scenario is shown in the video). The rotary sensor acts as a knob and allows precise movement of the door. This is also useful for debugging

Negative:

The system did feel quite cramped, so next time I would make it larger. This also opens the possibility of adding more complexity to the design, incorporating more sensors and actuators. I would have liked to add gestures and make use of the accelerometer sensor.

Another point, my ultrasonic sensor would constantly buzz and I could not find a way to properly fix it other than resetting the board. This is mentioned in my code (which includes debug prints for accessibility).

In conclusion, my mechatronics system has completed exactly what I set out to do, create a smart garage. It uses the required amount of sensors and actuators and works flawlessly. For next time, I would aim to make a more sophisticated sensor, perhaps incorporating more sensors and have a larger system for expandability, as my frame was quite cramped.